

Demand Forecasting with ARIMA, SARIMA, and Markov Chains: A Review of Fuel-Sales Forecasting Methods

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1 Introduction

Accurate demand forecasting is particularly important in supply-chain management because forecasts influence supply allocation, transportation planning, and the ability of firms to respond to changes in demand. Suppliers must predict future station requirements while coordinating inventory and transportation decisions across multiple stations.

The following paper is a review of the three forecasting methods applied in Wiecek and Kubek (2024) *The Impact of Time Series Selected Characteristics on the Fuel Demand Forecasting Effectiveness Based on Autoregressive Models and Markov Chains*: Autoregressive integrated moving average (ARIMA), seasonal autoregressive integrated moving average (SARIMA), and Markov-chain models.

The structure of Wiecek and Kubek (2024) is as follows: A brief literature review of existing research related to demand forecasting and supply-chain optimization, a fuel-demand case study using 85 stations in Poland, and a comparison of the different methods as applied to the fuel data.

2 Literature Review

The literature review in Wiecek and Kubek (2024) places fuel-demand forecasting within the broader context of supply-chain optimization. Across these applications, estimates of future demand can influence decisions involving inventory allocation, facility location, and transportation routing. A brief summary of some interesting and relevant use cases from this literature review is provided below.

Balaji et al. (2021) examine a demand-based warehouse relocation problem in which an optimization algorithm is used to minimize warehouse-related costs. Their approach incorporates several factors, including customer demand, the cost of serving customers, available supply, and the number of warehouses. The study illustrates how information about expected demand can be incorporated into larger logistical and facility-location decisions.

Moghdani et al. (2021) provide an overview of the Green Vehicle Routing Problem (GVRP), which can be seen as extension of the traditional Vehicle Routing Problem. The

GVRP considers transportation-routing decisions while incorporating environmental objectives such as reductions in CO_2 emissions. Although this problem is broader than fuel-demand forecasting itself, it reveals the importance of demand information in the optimization of transportation systems.

Fuel distribution presents a related forecasting problem because suppliers must determine how much fuel should be delivered to individual stations and how those deliveries should be incorporated into transportation routes. Fuel stations commonly operate under Vendor Managed Inventory (VMI) systems, in which suppliers receive current information from individual stations and assume responsibility for inventory-allocation decisions. Under such a system, improvements in demand forecasting may provide suppliers with better information for determining future fuel requirements and planning distribution activities. Taken together, the literature discussed in the article demonstrates that demand forecasting is closely connected to broader supply-chain decisions. This provides the motivation for comparing alternative forecasting methods for station-level fuel demand.

3 Data and Forecasting Methodology

3.1 Fuel-Demand Data

The data used to construct the forecasting models consist of daily fuel-sales volumes, measured in liters, from 85 fuel stations in Poland. The observations span the period from May 2018 through July 2022.

The individual time series exhibit substantial variation in daily fuel demand. The authors quantify the variability of each series using the ratio

$$\frac{\sigma}{\mu}, \tag{1}$$

where σ denotes the standard deviation of the series and μ denotes its mean. 58% of the time series had variability greater than 27%, 41% were below 27%, and 21% were greater than 50%, indicating that fuel demand varies considerably across the observed period.

The article also presents two example station-level time series to illustrate this variability. One of the selected series contains particularly large changes in demand over relatively short periods, including declines from several thousand liters per day to values near zero before increasing again. These fluctuations demonstrate some of the challenges associated with forecasting station-level fuel demand.

3.2 Rolling-Window Forecasting

The forecasting models are trained using the previous 180 observations of each fuel-demand series. The authors employ a rolling-window approach in which the estimation window is shifted forward by one observation for each subsequent forecast while maintaining a constant window length of 180 days.

As a result, the model is re-estimated for every forecast date using the most recent 180 observations. So, a window at time t is written as

$$\{Z_{t-179}, Z_{t-178}, \dots, Z_t\}, \quad (2)$$

and window shifted forward by one day is

$$\{Z_{t-178}, Z_{t-177}, \dots, Z_{t+1}\}. \quad (3)$$

This procedure allows model parameters to be updated as new demand observations become available, and observations older than 180 days are discarded.

3.3 ARIMA and SARIMA Forecasting

Before describing the specific model-selection procedure used by Wiecek and Kubek (2024), it is useful to briefly review the structure of the ARIMA family of models. ARIMA models describe the dependence of a time series on its own previous observations and on previous random disturbances. The general framework combines autoregressive (AR), differencing or integration (I), and moving-average (MA) components.

3.3.1 Autoregressive Models

An autoregressive model of order p , denoted $AR(p)$, represents the current value of a time series as a linear combination of its previous p values and a random error ε_t . Using Z_t to represent the observed time series, an $AR(p)$ process can be written as

$$Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \dots + \phi_p Z_{t-p} + \varepsilon_t, \quad (4)$$

where $\phi_1, \dots, \phi_p$ are the AR parameters.

Using the backshift operator B allows the series to be written much more compactly, defined as

$$BZ_t = Z_{t-1}, \quad (5)$$

and

$$B^k Z_t = Z_{t-k}. \quad (6)$$

As an example an $AR(2)$ model would be structured as

$$Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \varepsilon_t, \quad (7)$$

and rearranged to

$$Z_t - \phi_1 Z_{t-1} - \phi_2 Z_{t-2} = \varepsilon_t, \quad (8)$$

,

using the backshift operator, we get

$$(1 - \phi_1 B - \phi_2 B^2) Z_t = \varepsilon_t. \quad (9)$$

For an $AR(p)$ process, define the autoregressive polynomial

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p. \quad (10)$$

Then the AR(p) model can then be written simply

$$\phi(B)Z_t = \varepsilon_t. \quad (11)$$

3.3.2 Moving-Average Models

A moving-average model describes the current value of a time series using the current random error and errors from previous time periods. A moving-average model of order q , denoted MA(q), can be written as

$$Z_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q}, \quad (12)$$

where $\theta_1, \dots, \theta_q$ are the MA parameters.

Unlike an autoregressive model, which relates the current observation to previous observations of the series, the MA(q) model relates the current observation to current and previous random shocks. The order q therefore determines how many lagged errors are incorporated into the model.

Using the backshift operator the previous equation can be represented as

$$Z_t = (1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q) \varepsilon_t. \quad (13)$$

Defining the moving-average polynomial as

$$\theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q, \quad (14)$$

the MA(q) process can be written as

$$Z_t = \theta(B) \varepsilon_t. \quad (15)$$

Combining the autoregressive and moving-average components produces an ARMA(p, q) model. Using the autoregressive polynomial $\phi(B)$ defined previously, the ARMA model can be expressed as

$$\phi(B)Z_t = \theta(B)\varepsilon_t. \quad (16)$$

Thus, the autoregressive component describes dependence on previous values of the time series, while the moving-average component describes dependence on previous errors.

3.3.3 Stationarity and Differencing

A central requirement of ARMA modeling is stationarity. In a weakly stationary time series, the mean and variance are constant through time and the covariance between two observations depends only on the relative difference in lags rather than their absolute positions in the series. As usual,

$$E(Z_t) = \mu, \quad (17)$$

$$\text{Var}(Z_t) = \sigma^2, \quad (18)$$

$$\text{Cov}(Z_t, Z_{t-h}) = \gamma_h, \quad (19)$$

where μ and σ^2 do not depend on t and the autocovariance γ_h depends only on the lag h .

Stationarity is important because autoregressive models estimate temporal relationships under the assumption that the underlying statistical properties of the process remain sufficiently stable. If the mean, variance, or dependence structure changes systematically through time, relationships estimated from historical observations may not provide an appropriate description of future observations.

One common source of nonstationarity is a unit root. Consider the process

$$z_t = z_{t-1} + \varepsilon_t. \quad (20)$$

This process is a random walk. Because shocks to the process persist rather than gradually disappearing, its variance changes through time and the series is nonstationary.

Differencing can be used to remove this type of stochastic trend. The first difference of a series is defined as

$$\nabla Z_t = Z_t - Z_{t-1} \quad (21)$$

For the random walk,

$$\nabla Z_t = \varepsilon_t, \quad (22)$$

which is stationary.

The same differencing operation can also be represented using the backshift operator. Because

$$BZ_t = Z_{t-1}, \quad (23)$$

the first difference may be written as

$$\nabla Z_t = (1 - B)Z_t. \quad (24)$$

If additional differencing is required, the operation can be repeated. The d th difference is represented by

$$\nabla^d Z_t = (1 - B)^d Z_t. \quad (25)$$

The number of differences required to transform a series into a stationary series is represented by the integration parameter d .

3.3.4 ARIMA Models

An autoregressive integrated moving-average model, or ARIMA model, extends the ARMA framework by allowing the original series to be differenced before the autoregressive and moving-average components are applied. An ARIMA model is denoted

$$\text{ARIMA}(p, d, q), \quad (26)$$

where p is the order of the autoregressive component, d is the number of differences applied to the original series, and q is the order of the moving-average component.

If the differenced series is defined as

$$W_t = (1 - B)^d Z_t, \quad (27)$$

then an $\text{ARIMA}(p, d, q)$ model assumes that W_t follows an $\text{ARMA}(p, q)$ process:

$$\phi(B)W_t = \theta(B)\varepsilon_t. \quad (28)$$

Substituting the differenced series into this expression gives the general ARIMA representation

$$\phi(B)(1 - B)^d Z_t = \theta(B)\varepsilon_t \quad (29)$$

where

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p, \quad (30)$$

$$\theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q. \quad (31)$$

The integration parameter d is therefore directly related to the stationarity of the observed series. When $d = 0$, no differencing is required and the model reduces to an $\text{ARMA}(p, q)$ model. When $d = 1$, the first difference of the original series is modeled, while larger values of d correspond to repeated differencing.

3.3.5 Stationarity Testing with the KPSS Test

Because the degree of differencing d depends on whether the series is stationary, Wiecek and Kubek (2024) evaluate each fuel-demand series using the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test.

For the KPSS test, the hypotheses can be expressed as

$$H_0 : \text{the series is stationary}, \quad (32)$$

$$H_A : \text{the series is nonstationary}. \quad (33)$$

This direction of the hypotheses is important to note because the null is specified as the assumption that the series is stationary, opposed to most other unit-root tests where the null assumes the series is nonstationary.

In the study by Wiecek and Kubek (2024), the result of the KPSS test is used to determine the degree of integration required for the ARIMA model. Because the authors use a rolling 180-observation estimation window, the KPSS test is repeated as the window moves through the verification period.

Thus, the series can be classified as stationary during one slice of the window and nonstationary during another. The degree of integration can therefore respond to changes in the statistical behavior of each series as new observations enter the rolling window. This provides further motivation for repeatedly re-estimating the forecasting models rather than retaining a single model specification.

3.3.6 Seasonality and the SARIMA Model

Standard ARIMA models account for nonseasonal temporal dependence but do not explicitly represent repeating seasonal patterns. Fuel demand may exhibit such patterns because demand can vary systematically according to recurring daily, weekly, or seasonal behavior.

The Seasonal Autoregressive Integrated Moving Average model extends ARIMA by including additional seasonal autoregressive, differencing, and moving-average components. Using standard notation, the model is denoted

$$\text{SARIMA}(p, d, q)(P, D, Q)_m, \quad (34)$$

where

- p is the nonseasonal autoregressive order,
- d is the nonseasonal differencing order,
- q is the nonseasonal moving-average order,
- P is the seasonal autoregressive order,
- D is the seasonal differencing order,
- Q is the seasonal moving-average order, and
- m is the number of observations in one seasonal cycle.

For example, if daily observations followed a repeating weekly pattern, then $m = 7$ would represent a seven-day seasonal cycle. This example is illustrative; the seasonal period in the study was estimated separately from the observed data.

Seasonal differencing is analogous to ordinary differencing but compares observations separated by one complete seasonal period:

$$\Delta_m Z_t = Z_t - Z_{t-m}. \quad (35)$$

A SARIMA model can be represented compactly using the backshift operator as

$$\Phi(B^m)\phi(B)(1 - B)^d(1 - B^m)^D Z_t = \Theta(B^m)\theta(B)\varepsilon_t, \quad (36)$$

where $\Phi(B^m)$ and $\Theta(B^m)$ contain the seasonal autoregressive and moving-average components.

3.3.7 Model Identification and Selection

The authors construct separate ARIMA and SARIMA models for each of the 85 fuel-demand series. Each model is estimated using the most recent 180 observations. The final 30 observations of each series form the empirical verification period, during which forecasts are produced one day ahead.

Rather than fitting one model and using it throughout the verification period, the authors employ a rolling-window procedure. After each observed value in the verification period becomes available, it is added to the estimation sample while the oldest observation is removed. The sample therefore remains fixed at 180 observations, and both model parameters and model specifications may be re-estimated before the next one-step-ahead forecast is produced.

For the nonseasonal ARIMA parameters (p, d, q) , upper bounds $(p_{\max}, d_{\max}, q_{\max})$ are determined separately for each time series using the autocorrelation function (ACF) and partial autocorrelation function (PACF). These functions provide information about dependence between observations at different lags and can therefore assist in identifying appropriate autoregressive and moving-average orders.

For SARIMA models, the seasonal period m is determined using power spectral density analysis. The authors identify periodic components in the fuel-demand series and select the component that explains the greatest proportion of series variance as the seasonal period used in the model.

Within the resulting parameter bounds, different model specifications are evaluated iteratively, beginning with relatively simple specifications such as $(1, 1, 1)$ and progressing toward more complex alternatives. The preferred model is selected by minimizing the Akaike Information Criterion (AIC),

$$\text{AIC} = 2k - 2\ln(\hat{L}), \quad (37)$$

where k represents the number of estimated parameters and $\hat{L}$ represents the maximized likelihood of the fitted model.

The AIC introduces a tradeoff between model fit and model complexity. A more complex model may produce a larger likelihood, but the additional parameters increase the penalty term $2k$. Therefore, among the candidate specifications considered by the authors, the model with the smallest AIC is selected.

4 Markov-Chain Forecasting

In addition to ARIMA and SARIMA models, Wiecek and Kubek (2024) consider Markov chains as an alternative method for forecasting fuel demand. Unlike autoregressive models, which directly model numerical relationships between the lags, a Markov-chain model represents the process through a collection of states and the associated probabilities of transitioning between those states.

Let

$$X = (X_0, X_1, \dots, X_t, \dots, X_{t_{\max}}) \quad (38)$$

represent a sequence of random variables describing the state of the fuel-demand process through time (X_t is the state of the process at time t)

Let the finite state space be

$$S = \{s_1, s_2, \dots, s_k\} \quad (39)$$

with each s_i being one possible state of the process.

The sequence forms a Markov chain when the probability of the next state depends only on the current state and not the complete state history. So,

$$\begin{aligned} P(X_{t+1} = s_{t+1} \mid X_t = s_t, X_{t-1} = s_{t-1}, \dots, X_0 = s_0) \\ = P(X_{t+1} = s_{t+1} \mid X_t = s_t) \end{aligned} \quad (40)$$

This condition is known as the Markov property and provides information that conditional on the state of the process at time t , previous states do not provide any further information about the probability distribution of the state at time $t + 1$.

4.1 Transition Probabilities

The probabilities of moving between states is cleanly represented as a transition matrix. Let $\mathbf{P}$ denote the $k \times k$ transition matrix

$$\mathbf{P} = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1k} \\ p_{21} & p_{22} & \cdots & p_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ p_{k1} & p_{k2} & \cdots & p_{kk} \end{bmatrix} \quad (41)$$

Each element is defined as

$$p_{ij} = P(X_{t+1} = s_j \mid X_t = s_i) \quad (42)$$

So, p_{ij} is the conditional probability that the process moves from state s_i at time t to state s_j at time $t + 1$.

Because the elements of each row describe all possible transitions from a given state, the transition probabilities satisfy

$$p_{ij} \geq 0 \quad (43)$$

for all i and j , and

$$\sum_{j=1}^k p_{ij} = 1 \quad (44)$$

for each state s_i . The probability distribution across the state space at time t can be represented by the row vector

$$\boldsymbol{\pi}_t = [\pi_t(s_1) \quad \pi_t(s_2) \quad \cdots \quad \pi_t(s_k)] \quad (45)$$

For a homogeneous Markov chain, the same transition matrix is used at each time step, and the distribution at time $t + 1$ is

$$\boldsymbol{\pi}_{t+1} = \boldsymbol{\pi}_t \mathbf{P} \quad (46)$$

Repeating the process to an arbitrary time $t + h$:

$$\boldsymbol{\pi}_{t+h} = \boldsymbol{\pi}_t \mathbf{P}^h \quad (47)$$

A key assumption to use homogenous Markov chains is that the probabilities of transitioning between states is constant over time. Of course, this assumption would clearly be violated under most applications of fuel demand forecasting. Demand can be quite variable due to seasonality, changing consumer behavior, unpredictable shocks, and a myriad of other factors. Accordingly, Wiecek and Kubek (2024) use a non-homogenous Markov-chain model where the transition matrix is allowed to vary with time.

Let the transition matrix at forecast step τ be denoted by

$$\mathbf{P}(\tau) = \{p_{\tau,i,j}\}_{i,j \in S} \cdot \quad (48)$$

The state distribution can then evolve according to

$$\boldsymbol{\pi}_{t+\tau} = \boldsymbol{\pi}_t \mathbf{P}(\tau) \quad (49)$$

In this formulation, the Markov property is retained, but the probabilities of transitioning between states are allowed to depend on time. In the study, the verification horizon contains 30 observations, so

$$\tau \in \{1, \dots, 30\}. \quad (50)$$

This approach is useful for short-term forecasting because it does not require the transition structure of the demand process to remain fixed over the full forecasting period.

4.2 Rolling-Window Estimation

The non-homogeneous Markov-chain model is updated using the same general rolling-window principle applied to the ARIMA and SARIMA models. A transition matrix is estimated separately for each fuel-demand series using the current data window.

After a new observed demand value from the 30-day verification period becomes available, the rolling window is updated and the transition matrix is estimated again. A t_1 state distribution is then calculated using the updated transition probabilities.

The full forecasting process becomes

$$\text{Current data window} \rightarrow \mathbf{P}(\tau) \rightarrow \boldsymbol{\pi}_{t+1} \rightarrow \text{point forecast} \quad (51)$$

Allowing the transition matrix to be repeatedly re-estimated gives the Markov model the ability to adapt to changes in the observed transition behavior of fuel demand over time.

4.3 Construction of the State Space

To apply a Markov chain to numerical fuel-demand data, the continuous range of observed demand values must first be divided into a finite collection of states, so each state s_j is associated with an interval of fuel demand.

For each time series, Wiecek and Kubek (2024) determine the width of these demand intervals using

$$X_w = \frac{\max(X) - \min(X)}{\sqrt{N}}, \quad (52)$$

where $\max(X)$ and $\min(X)$ are the maximum and minimum observed values of the time series and N is the number of observations.

The resulting state space can be interpreted as a sequence of demand intervals of width X_w . For example, the first state corresponds to demand values beginning at $\min(X)$, the second state begins at $\min(X) + X_w$, and subsequent states continue across the observed range of the series.

Conceptually,

$$\begin{aligned} s_1 &: [\min(X), \min(X) + X_w), \\ s_2 &: [\min(X) + X_w, \min(X) + 2X_w), \\ &\vdots \\ s_k &: [\min(X) + (k-1)X_w, \max(X)] \end{aligned} \quad (53)$$

Each observed fuel-demand value can then be assigned to the corresponding state, allowing transition probabilities between demand ranges to be estimated.

The authors note that this interval-width rule is less standard than some commonly used methods for determining histogram classes. Their motivation is that the fuel-demand series used for the modeling is highly variable, and excessively narrow intervals could produce many states with few or no observed transitions.

4.4 Conversion to a Point Forecast

The Markov-chain model initially produces a probability distribution across possible demand states rather than a single numerical demand forecast. To compare the Markov-chain forecasts with the ARIMA and SARIMA forecasts, the predicted state distribution must therefore be converted into a point forecast.

Suppose the one-step-ahead state distribution is

$$\boldsymbol{\pi}_{t+1} = [\pi_{t+1}(s_1) \quad \pi_{t+1}(s_2) \quad \cdots \quad \pi_{t+1}(s_k)] \quad (54)$$

Let $\hat{X}_j$ denote the midpoint of the demand interval corresponding to state s_j . The point forecast is calculated as

$$Y_{t+1}^P = \sum_{j=1}^k \pi_{t+1}(s_j) \hat{X}_j \quad (55)$$

The resulting forecast is therefore a probability-weighted average of the midpoints of all possible demand states. States with greater predicted probability receive greater weight in the final numerical forecast.

For example, if the predicted state distribution places most of its probability on states representing higher levels of demand, the resulting point forecast will be shifted toward the midpoints of those higher-demand intervals. This conversion allows the probabilistic output of the Markov chain to be evaluated using the same forecast-error measures applied to the ARIMA and SARIMA models.

sectionForecast Evaluation

Forecast accuracy is evaluated over the 30-day verification period using mean absolute percentage error (MAPE) and root mean squared error (RMSE).

4.5 Mean Absolute Percentage Error

MAPE measures the average forecast error relative to the magnitude of the observed demand. It is defined as

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{Z_t - \hat{Z}_t}{Z_t} \right|, \quad (56)$$

where Z_t is the observed demand, $\hat{Z}_t$ is the corresponding forecast, and n is the number of observations in the verification period.

Because MAPE expresses forecast error as a percentage of observed demand, it allows forecasting performance to be compared across fuel stations with different demand levels.

4.6 Root Mean Squared Error

RMSE measures forecast error on the original scale of the response variable while placing greater weight on relatively large forecast errors. It is defined as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (Z_t - \hat{Z}_t)^2} \quad (57)$$

MAPE therefore emphasizes relative forecast accuracy, while RMSE is particularly sensitive to forecasts containing large absolute errors. Using both measures provides complementary information about model performance.

5 Forecasting Results

The Markov-chain and ARIMA models produced broadly comparable forecasting performance, while the SARIMA models exhibited larger errors and greater variability in performance across stations.

Table 1 summarizes the forecast-accuracy results reported in the article.

Table 1: Forecast accuracy comparison

Forecasting Method	Min. MAPE (%)	Median MAPE (%)	Max. MAPE (%)
ARIMA	2.06	16.75	72.12
SARIMA	10.06	22.35	120.36
Markov Chain	5.88	15.14	62.99

Forecasting Method	Min. RMSE	Median RMSE	Max. RMSE
ARIMA	42.22	207.94	902.02
SARIMA	45.85	208.33	890.51
Markov Chain	53.89	184.15	863.68

The Markov-chain model achieved the lowest median MAPE at 15.14%, compared with 16.75% for ARIMA and 22.35% for SARIMA. It also produced the lowest median RMSE at 184.15. These results suggest that the Markov-chain approach performed competitively with the autoregressive models across the collection of fuel-demand series.

ARIMA achieved the lowest minimum errors, including a minimum MAPE of 2.06% and minimum RMSE of 42.22. This indicates that ARIMA performed particularly well for some individual stations even though its median performance was slightly weaker than that of the Markov model.

SARIMA produced the highest median MAPE and the highest maximum MAPE, reaching 120.36%. Its median RMSE, however, was very similar to that of ARIMA. The difference between the MAPE and RMSE results show why multiple forecast-error measures are useful: conclusions about relative model performance can depend on whether errors are evaluated in absolute units or relative to the magnitude of demand.

Overall, the reported results do not indicate that one forecasting method dominates for every fuel-demand series. Instead, forecast effectiveness appears to vary across stations, motivating the authors' subsequent examination of whether characteristics of the individual time series are associated with forecasting performance.

6 Time-Series Characteristics and Forecast Error

In addition to comparing overall forecasting accuracy, Wiecek and Kubek (2024) examine whether characteristics of the individual fuel-demand series are associated with model performance. This analysis is intended to determine whether certain types of demand behavior are systematically easier or more difficult to forecast.

The evaluated time-series characteristics include

- variance,
- standard deviation,
- power spectral density,

- predominant cycle length,
- Hurst exponent, and
- first-lag autocorrelation.

For each characteristic, the authors calculate Pearson correlation coefficients with MAPE. Statistical significance is evaluated using two-sided tests at the significance level

$$\alpha = 0.05. \tag{58}$$

A positive correlation indicates that larger values of a given time-series characteristic are associated with larger percentage forecast errors, while a negative correlation indicates an association with lower forecast errors.

The strongest and most consistent relationship across the three forecasting methods is observed for predominant cycle length. Its correlation with MAPE is -0.51 for SARIMA, -0.59 for the Markov-chain model, and -0.51 for ARIMA. All three correlations are statistically significant. Longer predominant cycles are therefore associated with lower forecast errors across all three methods, with the strongest relationship occurring for the Markov-chain forecasts.

Variance is positively and significantly associated with MAPE for all three forecasting methods. The correlations are 0.52 for SARIMA, 0.30 for the Markov-chain model, and 0.22 for ARIMA. This indicates that fuel-demand series with greater variability tend to be associated with larger percentage forecast errors.

Standard deviation follows a similar positive pattern, with correlations of 0.47 , 0.21 , and 0.18 for SARIMA, the Markov-chain model, and ARIMA, respectively. However, only the SARIMA correlation is statistically significant. Similarly, power spectral density has correlations of 0.35 for SARIMA, 0.21 for the Markov-chain model, and 0.11 for ARIMA, with only the SARIMA relationship reaching statistical significance.

First-lag autocorrelation is positively associated with MAPE for all three models, with correlations of 0.43 for SARIMA, 0.24 for the Markov-chain model, and 0.31 for ARIMA. These relationships are statistically significant, indicating that greater dependence between consecutive demand observations is associated with larger percentage forecast errors in the analyzed series.

In contrast, the Hurst exponent shows essentially no linear relationship with forecast accuracy. Its correlations with MAPE are -0.06 for SARIMA, -0.06 for the Markov-chain model, and -0.08 for ARIMA, and none of these correlations are statistically significant.

Overall, the results indicate that forecast accuracy depends not only on the forecasting method but also on characteristics of the underlying demand series. Predominant cycle length shows the strongest consistent relationship with forecast error, while greater variance is associated with poorer forecasting performance. SARIMA also appears more sensitive to standard deviation and power spectral density than either ARIMA or the Markov-chain model. In contrast, the Hurst exponent provides little evidence of a relationship with forecasting accuracy in the analyzed data.

7 Discussion

The results reported by Wiecek and Kubek (2024) show that forecasting performance varies across both forecasting methods and individual fuel-demand series. The Markov-chain model achieved the lowest median MAPE and RMSE, while ARIMA produced the lowest minimum errors. These results suggest that no single method is uniformly superior across all stations.

The strong performance of the Markov-chain approach is notable because it models demand differently from ARIMA and SARIMA. Rather than directly modeling numerical relationships among previous observations, it divides demand into discrete states and forecasts future demand using estimated transition probabilities. Despite this simplification, the Markov-chain model produced the lowest median errors, suggesting that its rolling, non-homogeneous transition structure can adapt effectively to changing demand behavior.

SARIMA produced less favorable results. Although it explicitly incorporates seasonal structure, it had the highest median MAPE and substantially larger maximum percentage errors. This suggests that additional model complexity does not necessarily improve forecast accuracy, particularly when seasonal patterns are unstable or must be estimated from a relatively short 180-observation rolling window.

The analysis of time-series characteristics further indicates that forecast performance depends on the structure of the underlying demand series. Predominant cycle length showed the strongest consistent relationship with MAPE, with longer cycles associated with lower forecast errors across all three methods. Greater variance was associated with higher MAPE, indicating that more variable demand series tend to be more difficult to forecast. The Hurst exponent, in contrast, showed little relationship with forecasting accuracy.

An important strength of the study is its rolling-window estimation procedure, which allows models to incorporate new observations as they become available rather than applying one fixed specification throughout the verification period. However, the analysis also has limitations. The data are limited to 85 fuel stations in Poland, and forecast performance is evaluated over only 30 observations per series. In addition, the models rely primarily on historical demand and do not incorporate external variables such as weather, fuel prices, holidays, or economic conditions.

The study demonstrates that forecasting effectiveness depends on both model choice and the characteristics of the series being forecast. The results support comparing alternative forecasting approaches empirically rather than assuming that a more complex model will necessarily provide better forecasts.

8 Conclusion

The study reviewed in this paper demonstrates that demand-forecasting performance depends on both the forecasting method and the characteristics of the underlying time series. Markov-chain and ARIMA models generally performed better than SARIMA, while measures such as variance and predominant cycle length were associated with forecast accuracy. Overall, the results emphasize the value of comparing multiple forecasting approaches and adapting model selection to the behavior of the individual demand series.

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